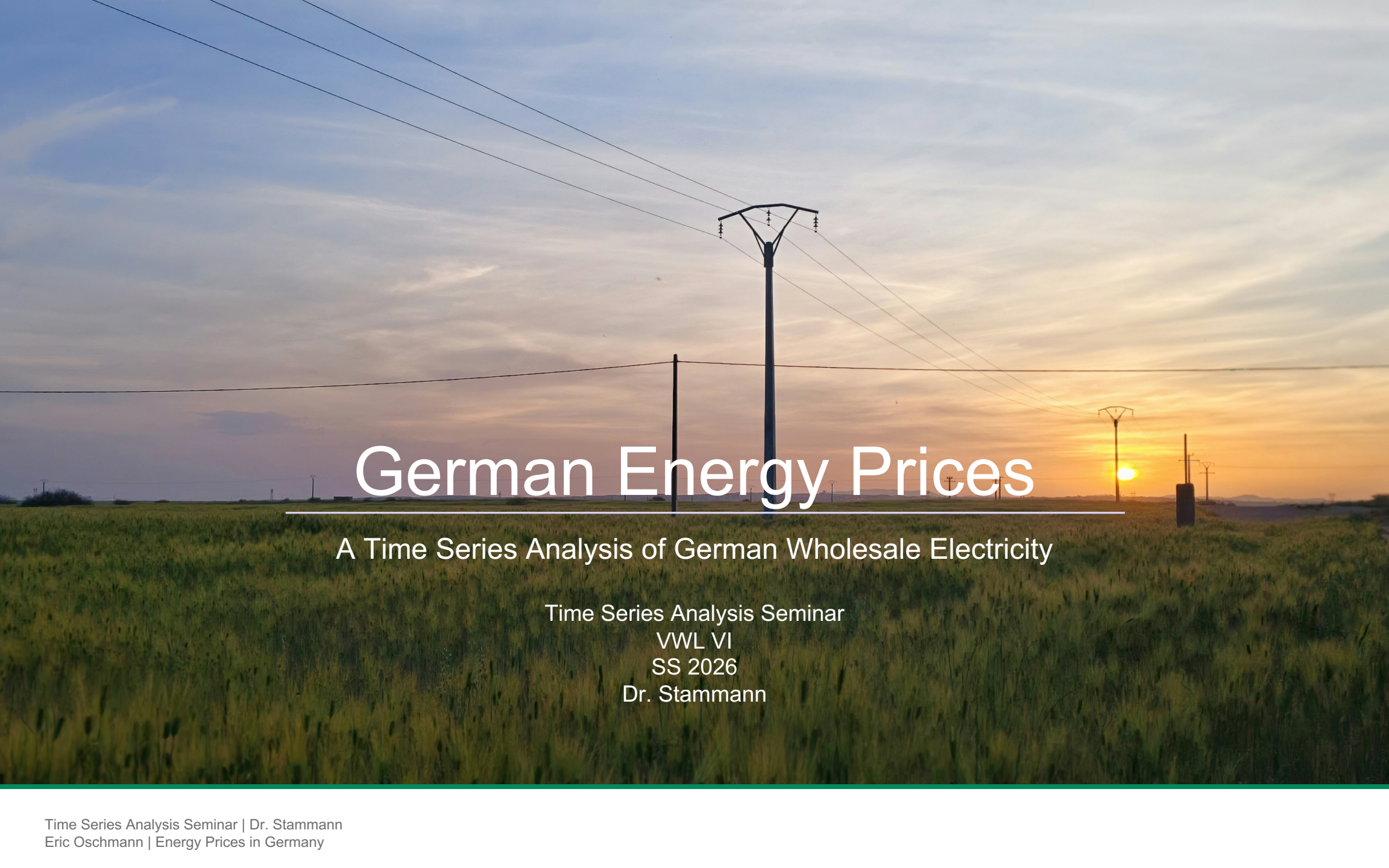




German Energy Prices

A Time Series Analysis of German Wholesale Electricity

Time Series Analysis Seminar
VWL VI
SS 2026
Dr. Stammann

Economic Motivation

- Households and companies are affected by energy prices
 - ➔ Effect on daily lives
- Some investment decisions rely on assumptions about energy prices

Germany worries about gas rationing as supply from Russia halted

Temporary closure of Nord Stream 1 prompts fears for private consumers as well as industry

Source: Oltermann, P. (2022). "Germany worries about gas rationing as supply from Russia halted." *The Guardian*. <https://www.theguardian.com/world/2022/jul/19/germany-worries-about-gas-rationing-as-supply-from-russia-halted>

Personal Motivation

- Current events motivate discussion about reliance on global energy supplies
 - ➔ Geopolitical leverage through dependence on energy prices
- Unpredictable energy prices **may drive renewable energy adoption**

How Europeans Are Responding to Exorbitant Gas and Power Bills

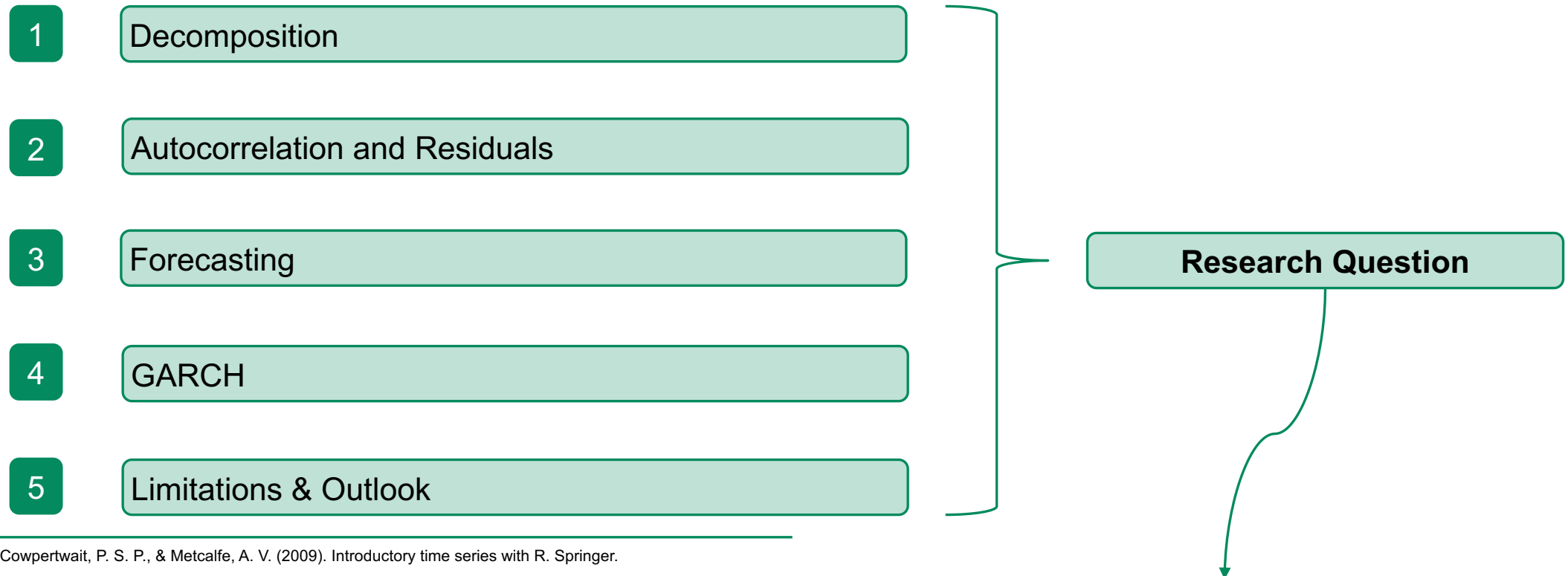
High energy prices in Europe are upending people's lives. While some are installing solar panels, others are stoking their wood-burning stoves.

Source: Eddy, M., & Alderman, L. (2022). "How Europeans Are Responding to Exorbitant Gas and Power Bills." *The New York Times*. https://www.nytimes.com/issue/todayspaper/2022/02/22/todays-new-york-times?eafs_enabled=false

Methodology

- Time Series Analysis in R: German/Luxemburg Wholesale Energy Price Series
- Methodology based on Metcalfe & Cowpertwait Book on Time Series Analysis.¹

Structure



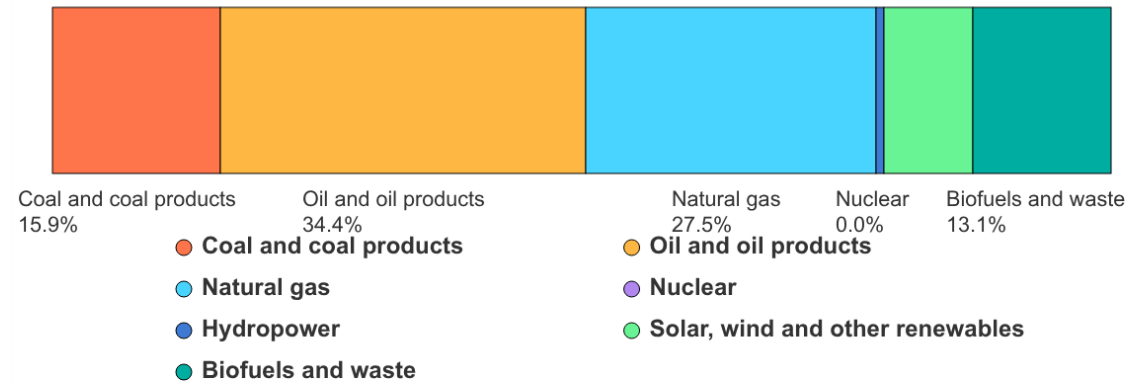
¹ Cowpertwait, P. S. P., & Metcalfe, A. V. (2009). Introductory time series with R. Springer.

Does including the 2022 energy crisis period degrade forecast accuracy of energy prices and does the same hold for countries relying on different primary energy consumption patterns?

Energy in Germany

- Prior to February 24th, 2022
➔ Reliance on natural gas imports from Russia
- Russian Invasion fueled energy price crisis
- Similar energy price movements & crisis in other European countries

Total energy supply, Germany, 2024



Source: International Energy Agency. Licence: CC BY 4.0

Total Yearly German Energy Demand:
476.000.000 MWh¹

Average Consumption:
3.5 MWh/ year
2-Person Household

Average Yearly Costs:
1150 – 1450 €/ year
2-Person Household

```
> cor(weeklyDE, weekly$`Frankreich [€/MWh]`)  
[1] 0.9647013
```

```
> cor(weeklyDE, weekly$`Österreich [€/MWh]`)  
[1] 0.9897348
```

```
> cor(weeklyDE, weekly$`Norwegen 2 [€/MWh]`)  
[1] 0.9529006
```

Graph: International Energy Agency. Germany: Energy Mix. <https://www.iea.org/countries/germany/energy-mix>, accessed July 4, 2026.

¹ ibid.

Data Source

- SMARD Electricity and Gas Market Data ¹
 - ➔ Federal Bureau for Grids and Energy Systems (Bundesnetzagentur)
- Provides granular energy price data for monthly, weekly, hourly and 15-minute intervals
- German and Luxembourg Wholesale Prices per MWh [€/MWh]

```
> head(weekly[ "Deutschland/Luxemburg [€/MWh]"), 3)
```

```
`Deutschland/Luxemburg [€/MWh]`  
1          51.4  
2          54.5  
3           64
```



NA's
0

Observations
379

Min.	Median	Mean	Max.
11.19	73.85	93.23	585.92



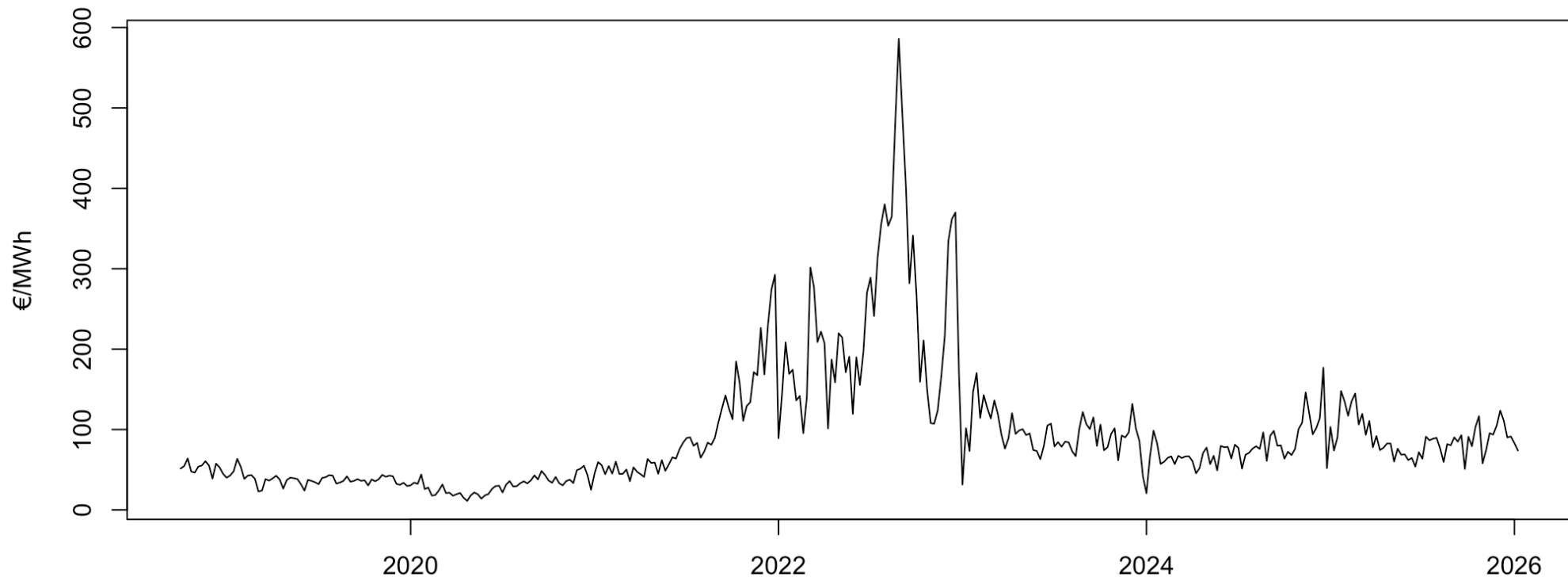
¹ Bundesnetzagentur. *SMARD: Electricity and gas market data.*

² Statistisches Bundesamt. (2023). *Stromverbrauch der privaten Haushalte nach Haushaltsgrößenklassen.*

Wholesale Energy Price Series

- Energy Price Crisis after Russian invasion of Ukraine

German Weekly Wholesale Energy Prices



Time
1.October 2018 - 05.January 2026

Assumptions

Time Series starting 1.10.2018

Weekly Price Data

German **Wholesale** Prices

Frequency = 52

Log transformed prices

Reasoning

Change of Price Regions from GER/LUX/AUS to GER/LUX¹

Avoids daily noise and preserves visibility of seasonality

Household energy prices depend on provider

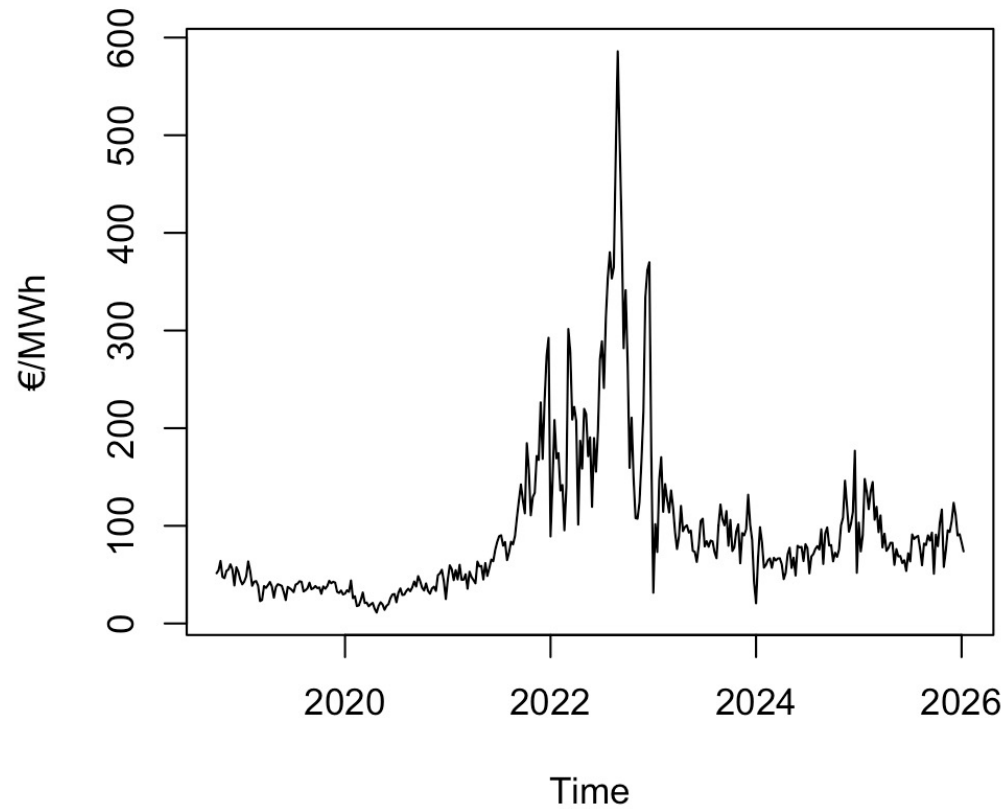
Simplification, instead of approx. 52.18 weeks/year

Stabilises variance and normalizes large spikes & differences

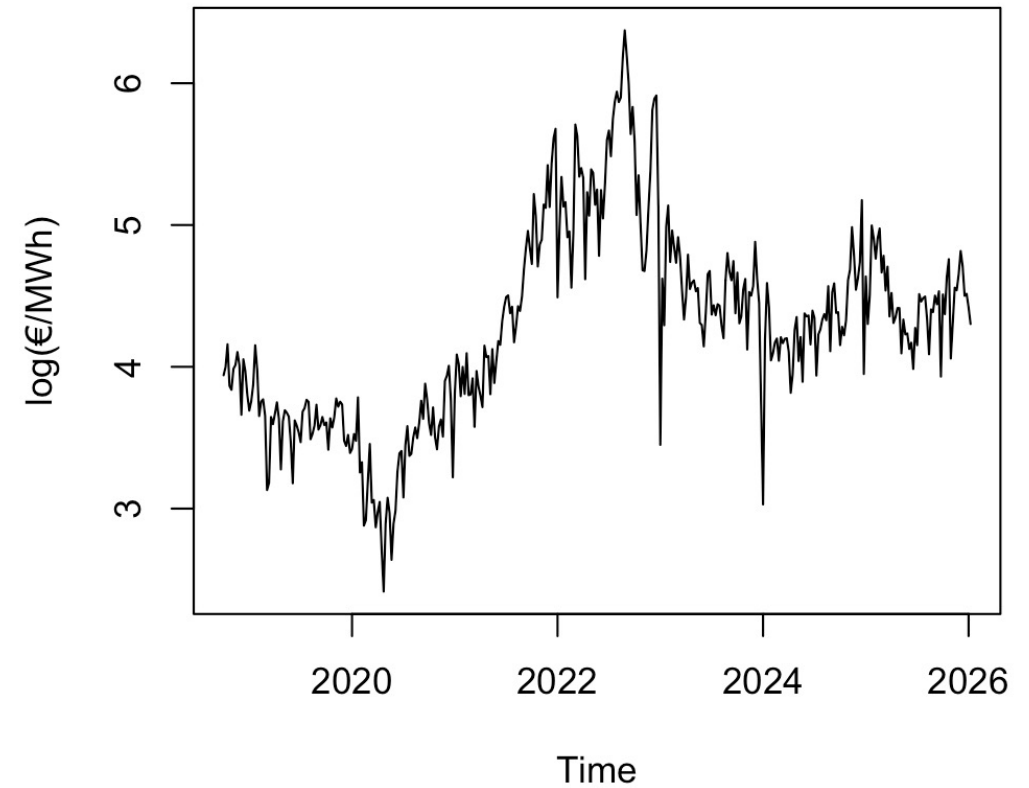
² Bundesnetzagentur. Engpassbewirtschaftung an der Grenze zu Österreich. SMARD.

Log-Transformed Series Comparison

Level series



Log series



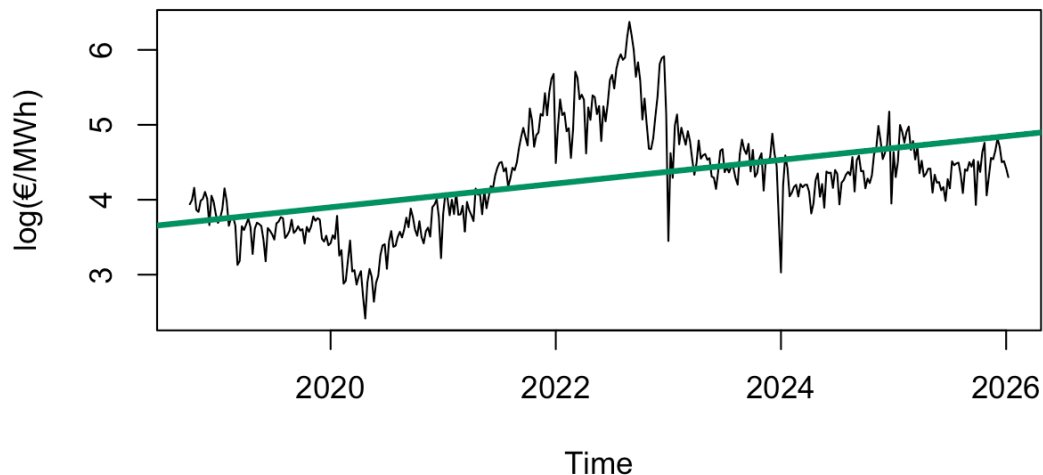
Theoretical Interpretation

- C.p. every year electricity price increases by 17.2 %.

Limitation

- Slope is driven by crisis period after Russian invasion
- Lacks explanatory power and forecasting quality

Log weekly prices with linear trend



=====	
Dependent variable:	

log(Weekly electricity price)	

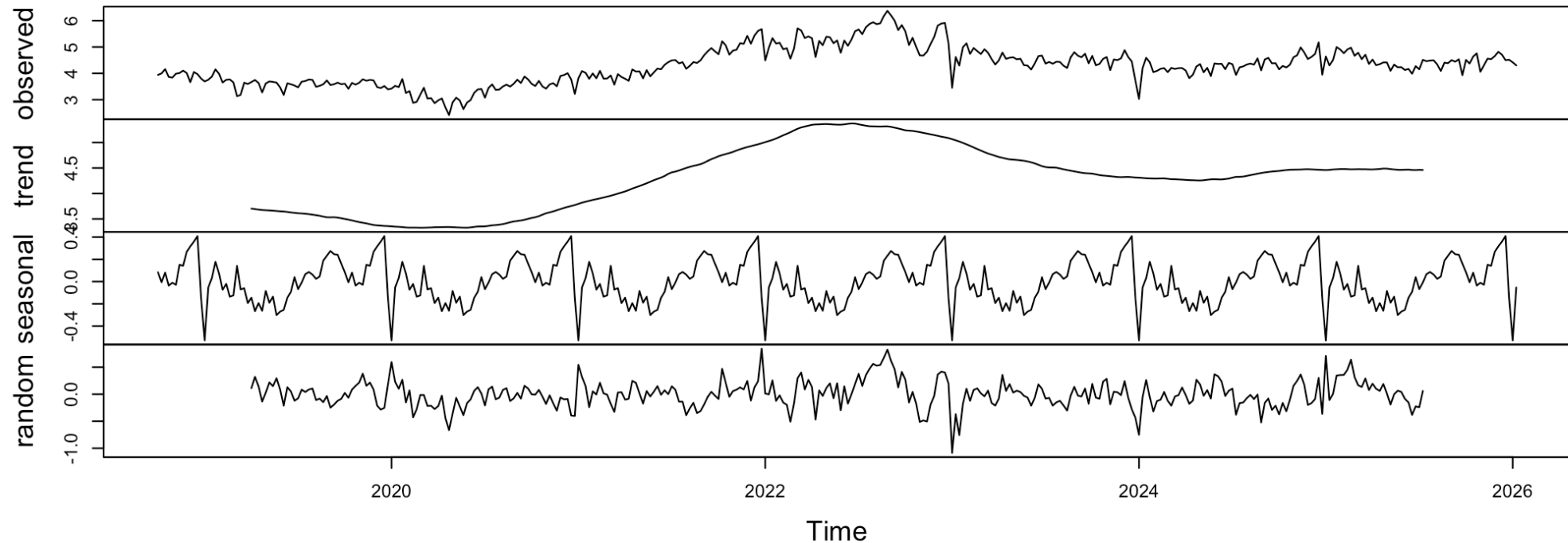
Time	0.15840 (0.01494) p = 0.00000
Constant	-316.06668 (30.20782) p = 0.000

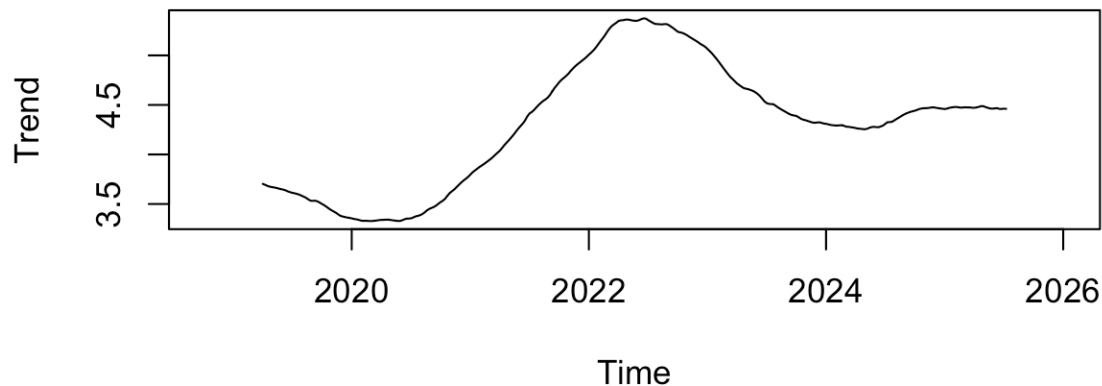
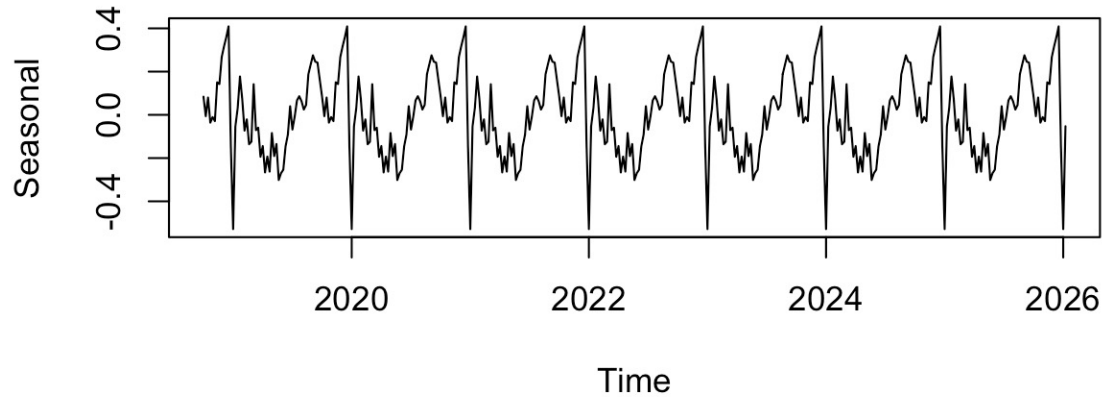
Observations	379
R2	0.2298
Adjusted R2	0.2277
=====	
Note:	*p<0.1; **p<0.05; ***p<0.01

Additive Decomposition: $\log(x_t) = m_t + s_t + z_t$

where, at time t , $\log(x_t)$ is the log-transformed energy price series, m_t is the trend, s_t is the seasonal effect, and z_t is an error term that is a sequence of correlated random variables with mean zero.

Decomposition of additive time series





Seasonal Component

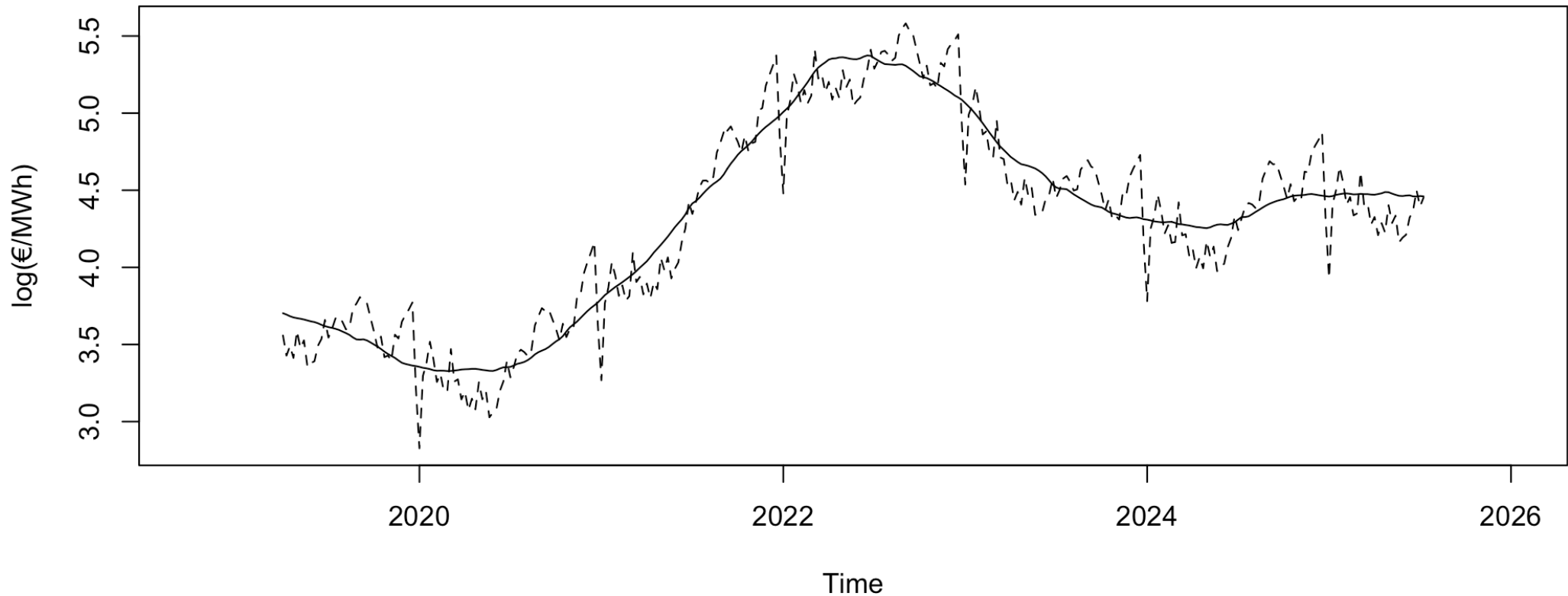
- Presence of annual recurring patterns
- ➔ winter peak demand, summer low demand
- ➔ renewable generation cycles
- ➔ energy system-stress periods

Trend Component

- Clear trend distortion during energy price crisis as a result of the Russian invasion of Ukraine.
- Pre-/post-crisis levels far below crisis peak

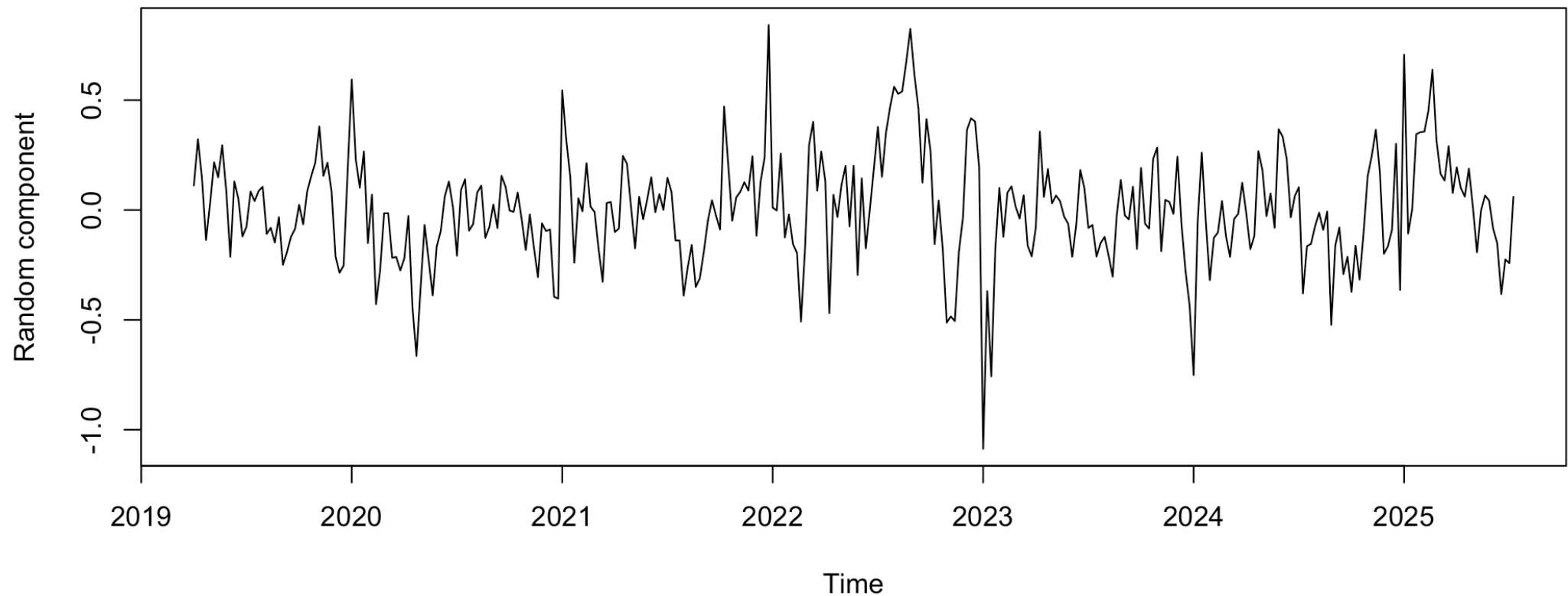
Trend and Trend + Seasonal Plot

- Seasonal effect is small relative to the trend



Random Component

- Spikes during German energy crisis after Russian invasion



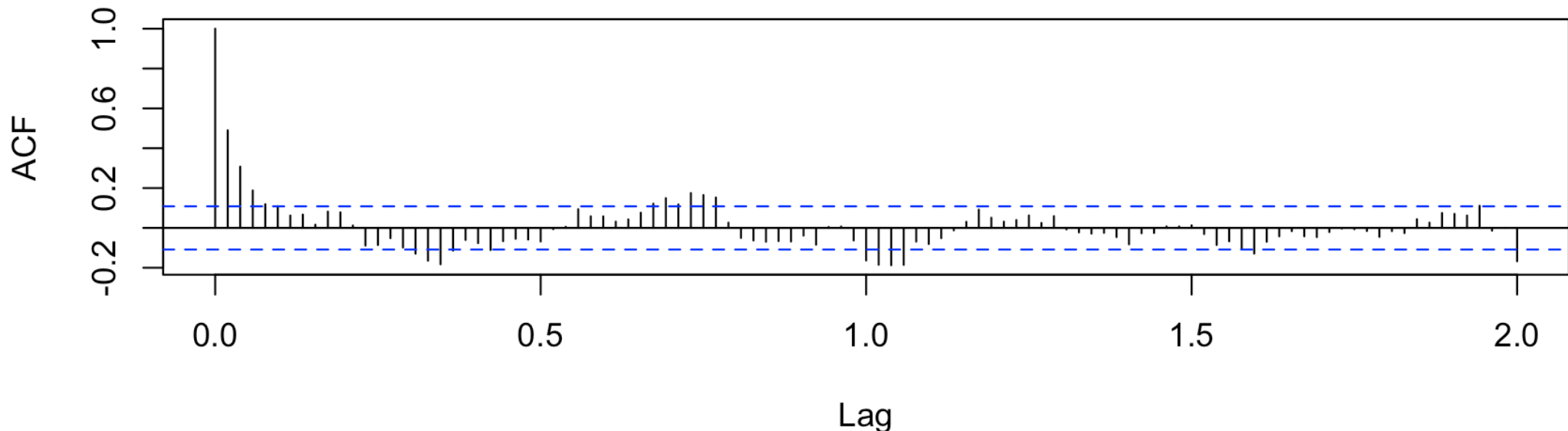
Autocorrelation in Random Component

- Not white noise, autocorrelation at lag 1 and 2
- ➔ Weekly, annual autocorrelation
- Short-run weekly autocorrelation

```
> #Autocorrelation current week and previous week: [2]  
> acf(weekly.random)$acf[2]  
[1] 0.4901079
```

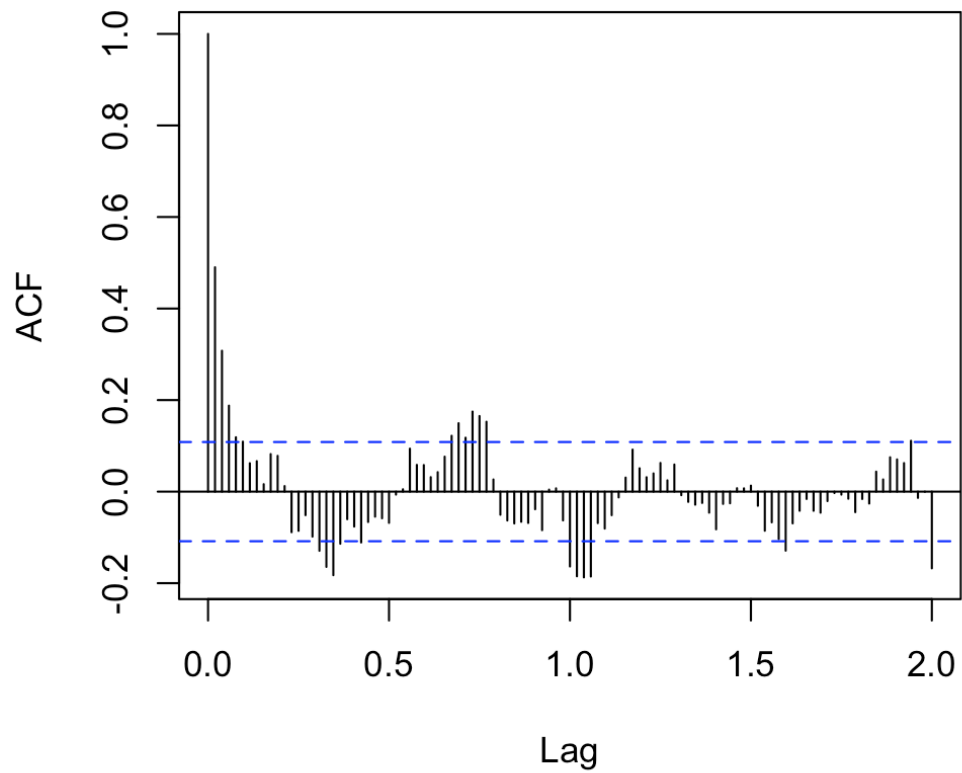
```
> #Autocorrelation current week and one month earlier: [5]  
> acf(weekly.random)$acf[5]  
[1] 0.1187632
```

Series weekly.random

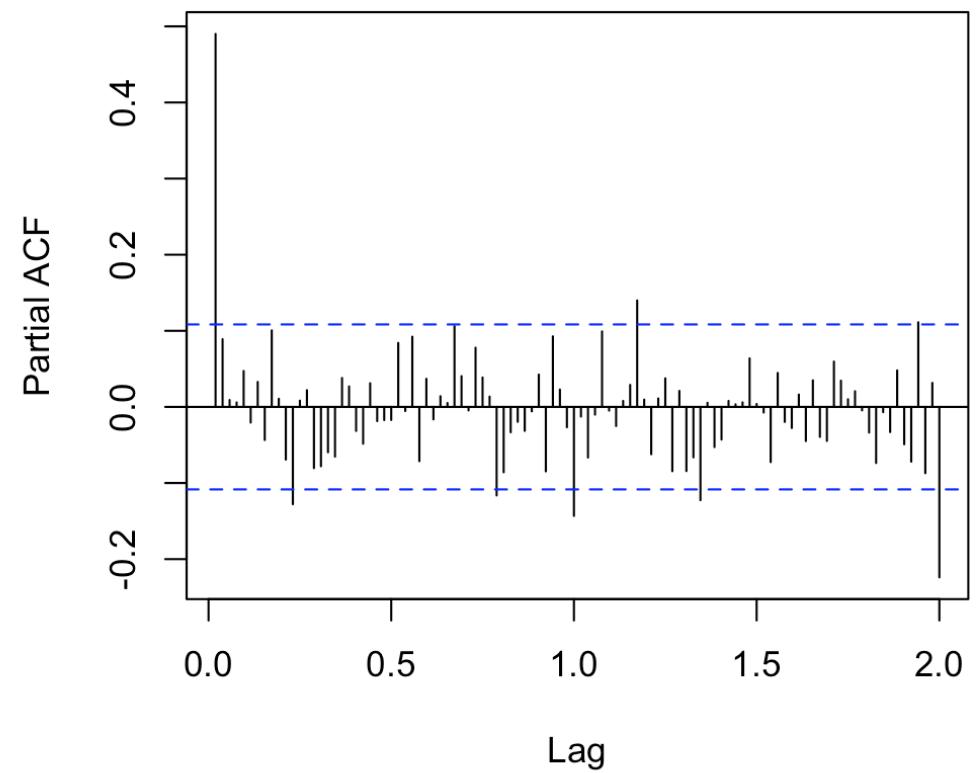


ACF and PACF Overview

ACF: random component



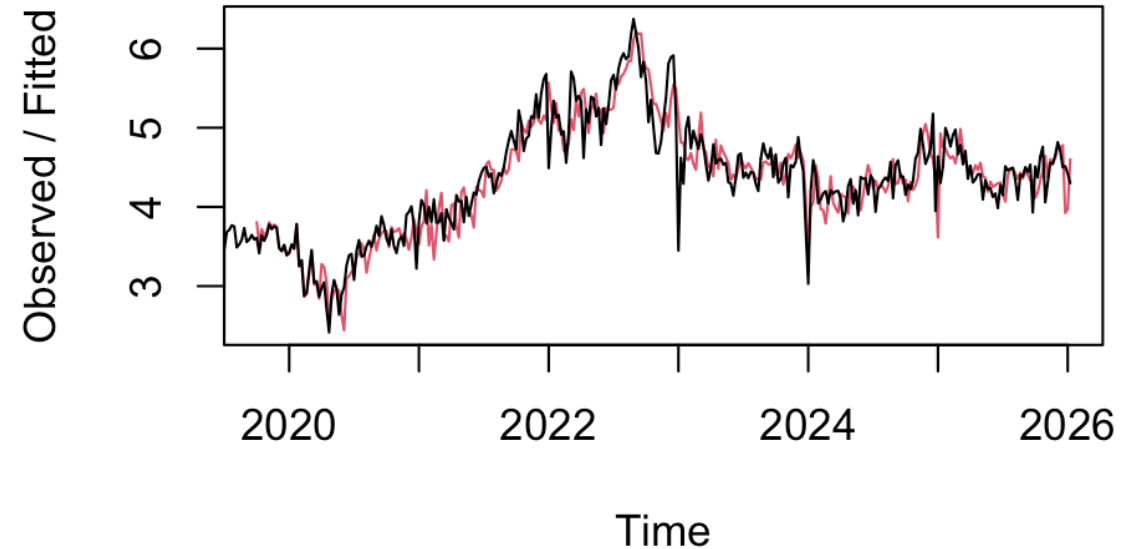
PACF: random component



Holt-Winters Approach

- Baseline estimation model
- ➔ Unlikely to fit given large trend component in series
- **Key Problem of the series:**
- ➔ Crisis trend influence make forecasting more complex

Holt-Winters filtering



Preliminary Insights

- Weekly price remain relevant
- Annual weekly seasonality

Call:
`HoltWinters(x = weekly.ts, seasonal = "additive")`

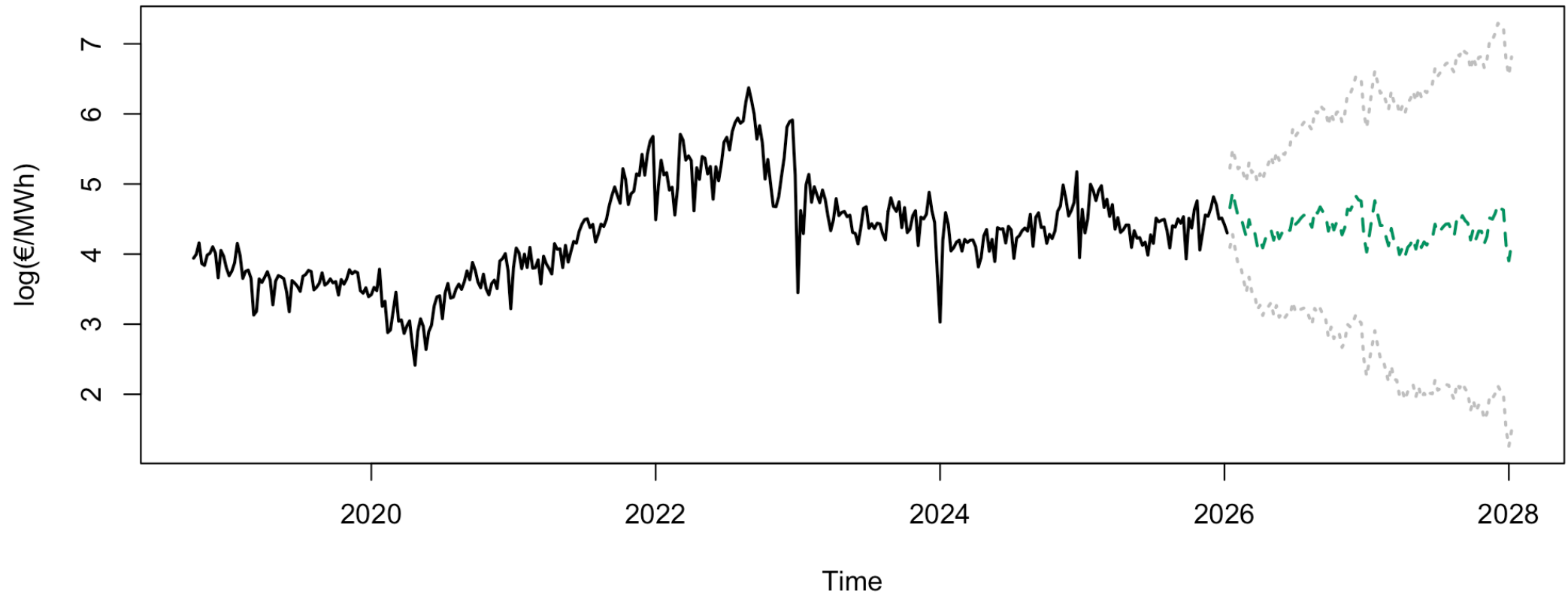
Smoothing parameters:

alpha: 0.3914396

beta : 0.002800409

gamma: 0.4574828

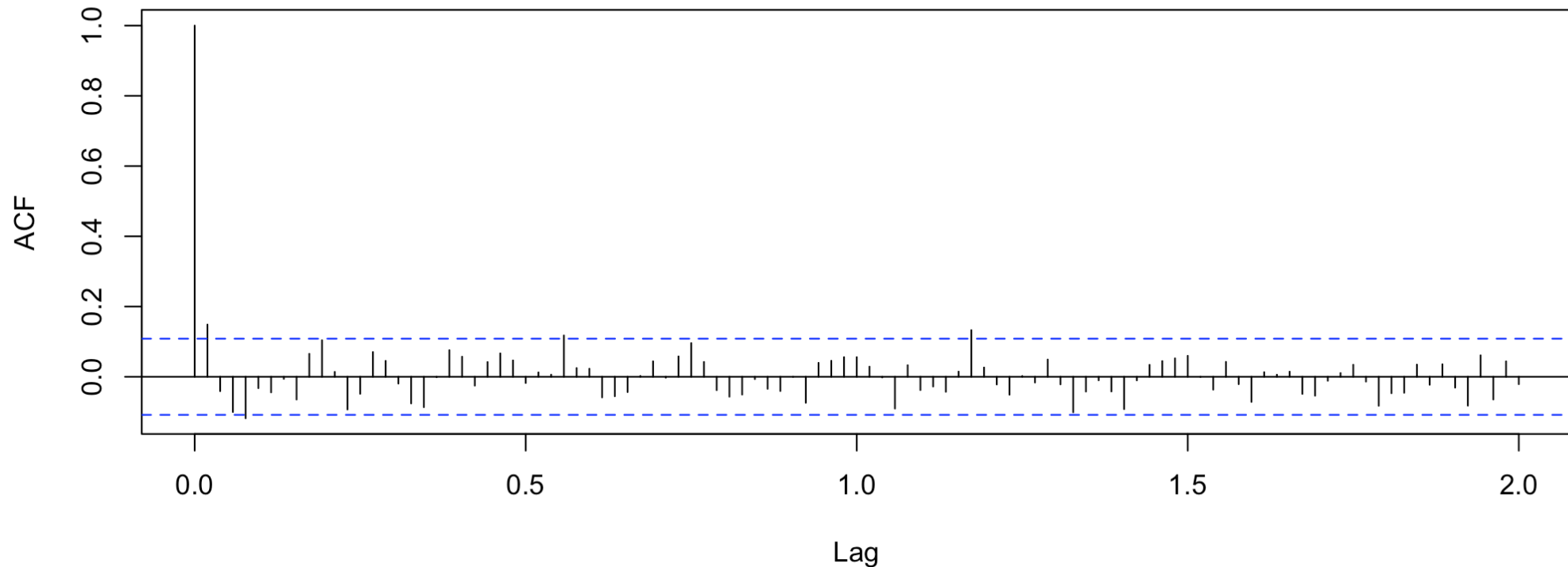
Holt-Winter 2 Year Prediction



ACF of Holt-Winters residuals

- Not completely white noise structure
- ➔ Some autocorrelation remains in Holt-Winters residuals

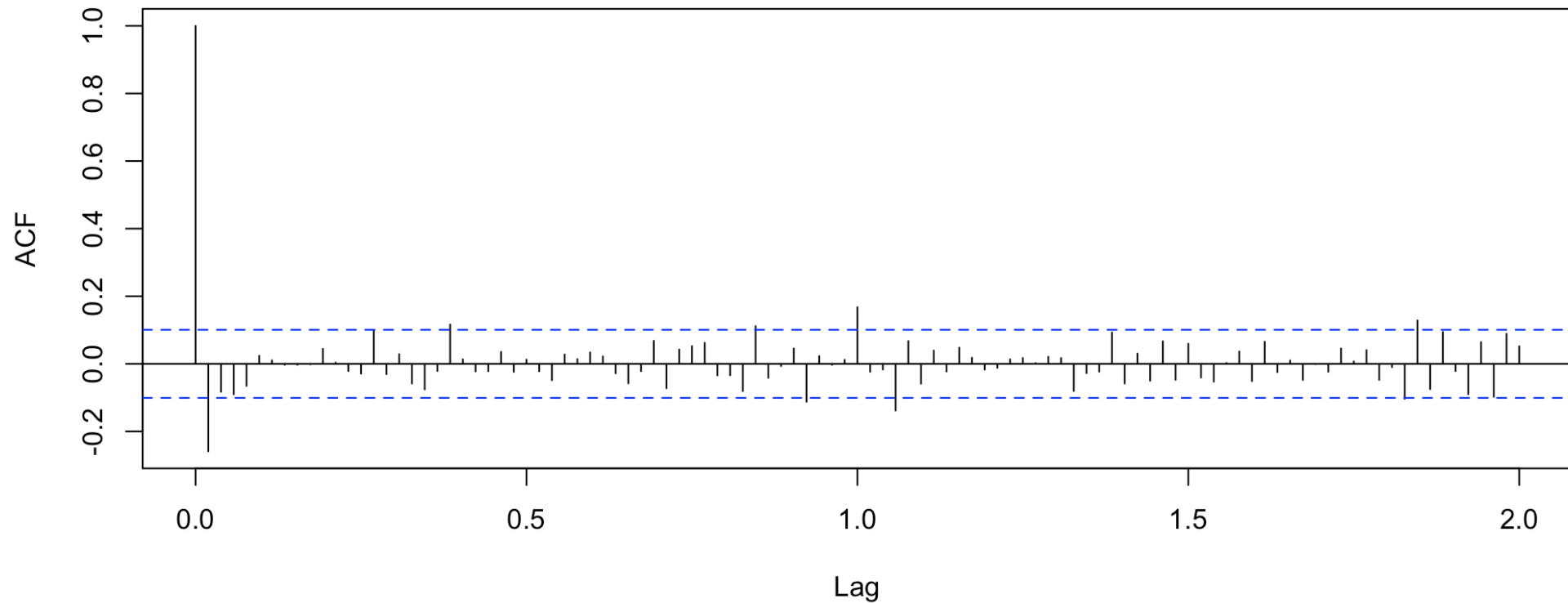
ACF: Holt-Winters residuals



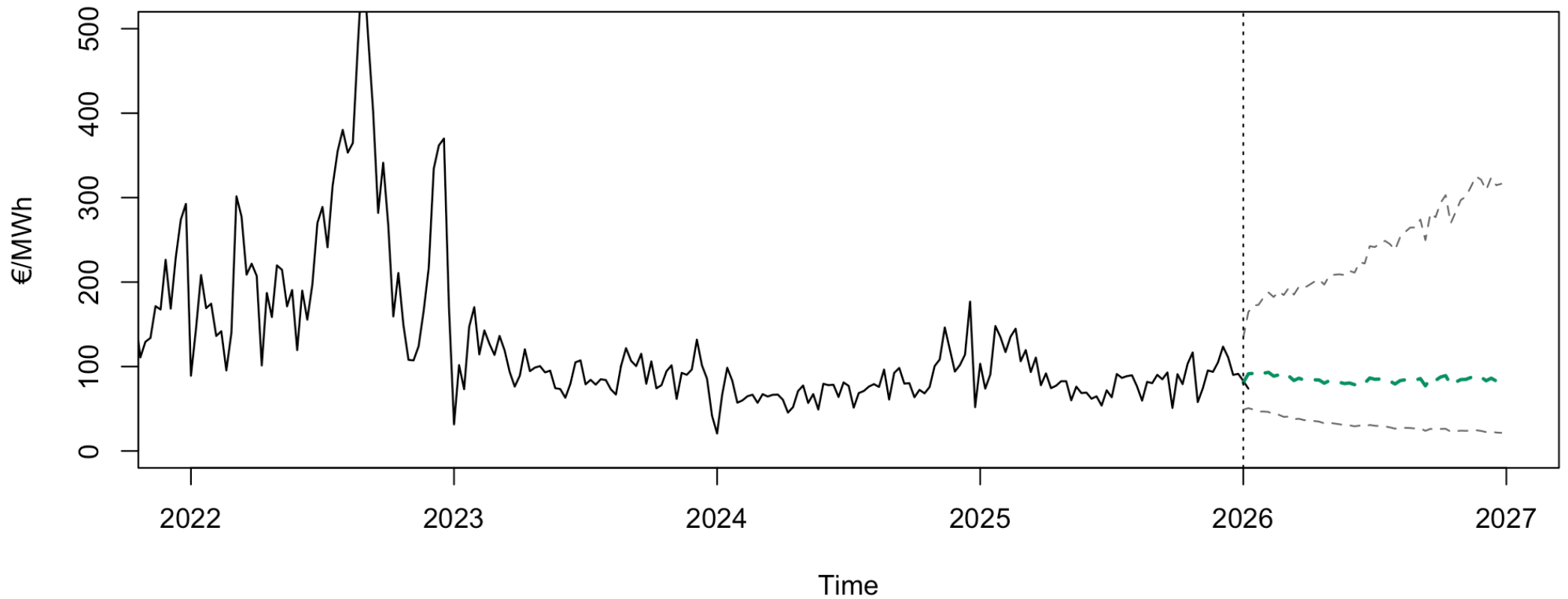
Checking Suitability: Random Walk

- Little autocorrelation in differenced series
- ➔ Closer to white noise

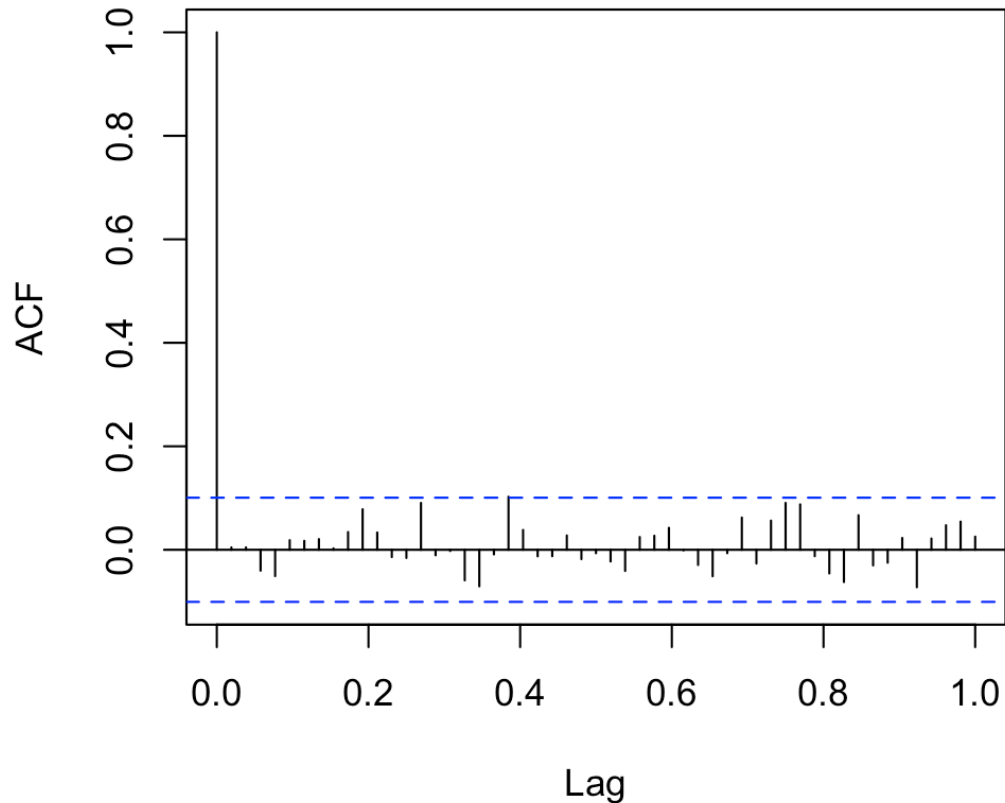
ACF: diff(weekly.ln)



SARIMA (1,1,1)(0,0,1) Forecast

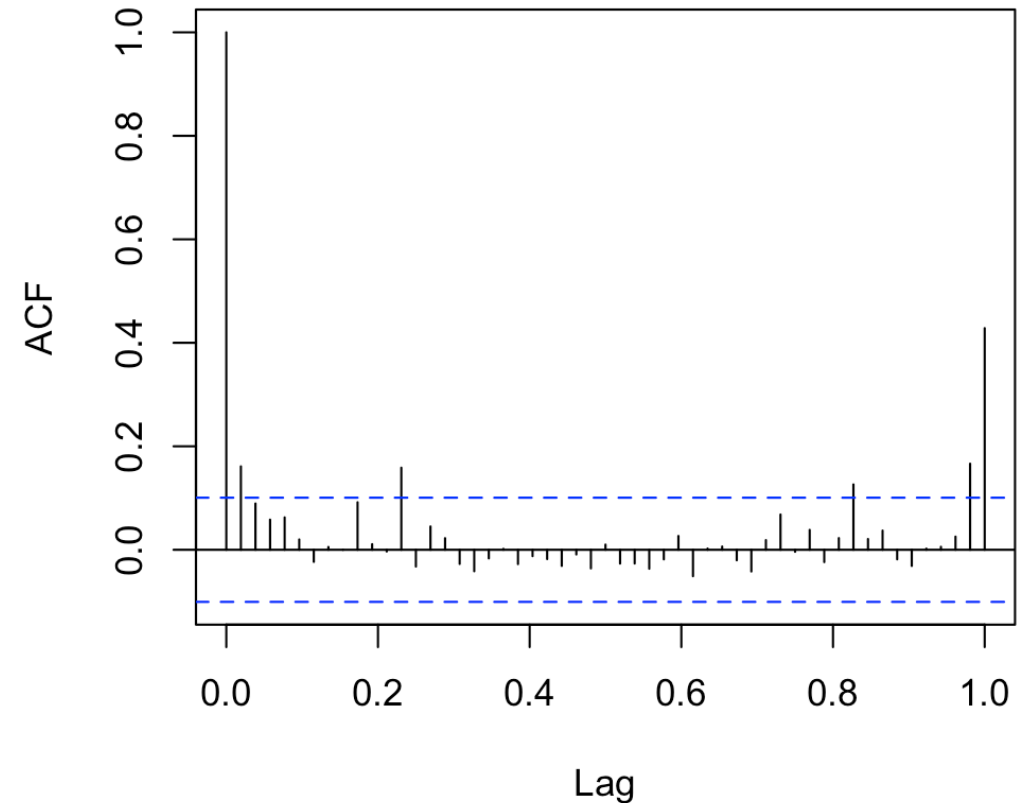


ACF: SARIMA residuals



- No significant autocorrelation remains

ACF: Squared SARIMA residuals



- Significant autocorrelation at short lags
➔ Volatility clustering

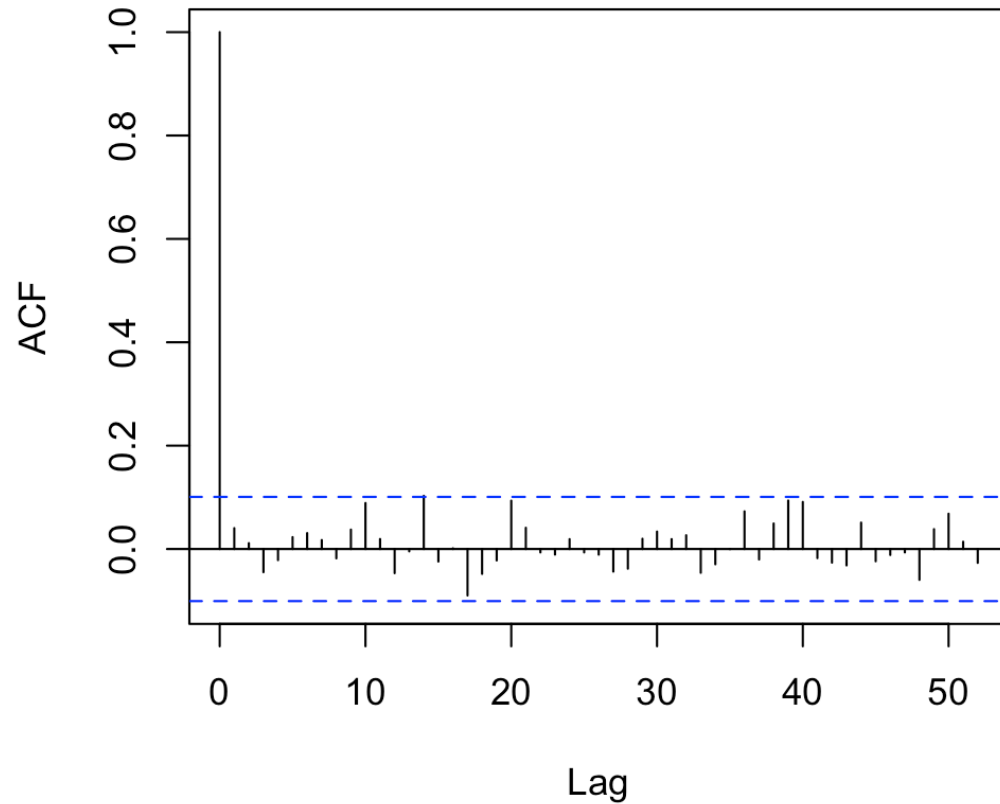
GARCH Model

- Fitting GARCH on residuals of SARIMA
- **a1** significant
→ Volatility clustering
- **b1** confidence interval crosses zero
→ No evidence of persistent volatility shocks
- Volatility reacts strongly to shocks but does not persist
- Short volatility spikes as a result of energy crisis instead of lasting „high-volatility regime“

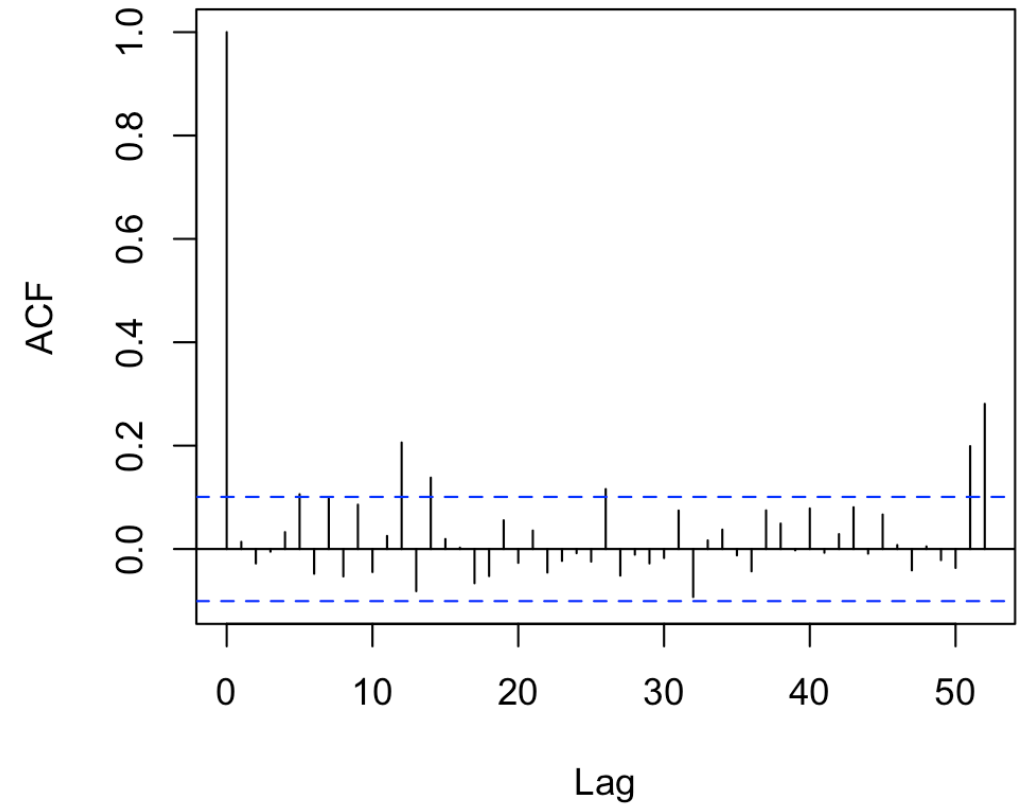
```
> confint(weekly.garch)
```

	2.5 %	97.5 %
a0	0.01296059	0.04727727
a1	0.22812916	0.44726687
b1	-0.08192586	0.52336389

ACF: GARCH residuals



ACF: Squared GARCH residuals



Limitations

- Series limited to 379 observations
 - ➔ No current data of price and volatility spike

- Choice of forecast training window
 - ➔ Forecast testing window has effect on accuracy and results of the forecast

- Annual volatility spikes remain
 - ➔ Long Memory Processes from Book, Chapter 8

- Continuous seasonal autocorrelation at lag = 52
 - ➔ ACF of Squared GARCH residuals
 - ➔ Add Seasonal GARCH or P-GARCH

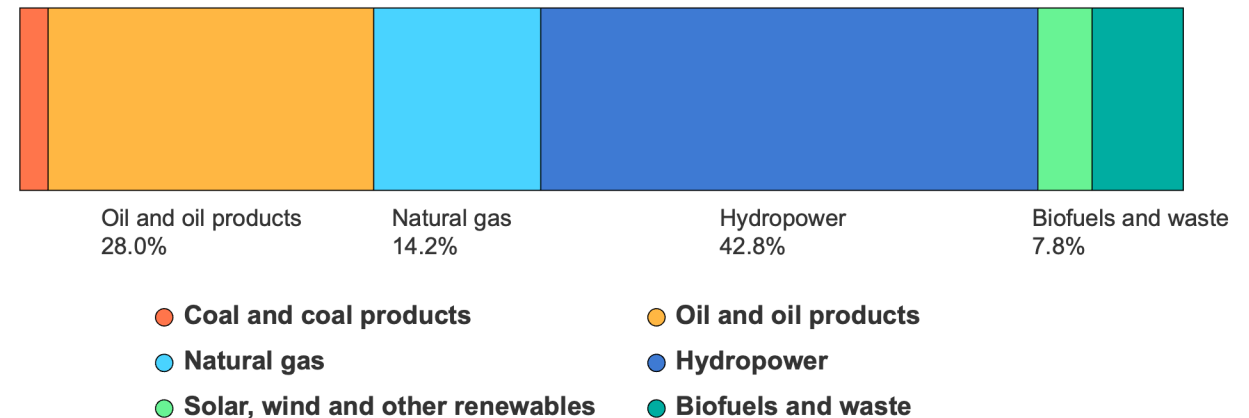
Next Steps

- Compare with countries with higher renewable share and different primary energy consumption mix

Norway

- Largest share of hydropower generation
- 42.8% Hydropower generation
- Different seasonal component structure due to energy consumption patterns

Total energy supply, Norway, 2024



Source: International Energy Agency. Licence: CC BY 4.0

Do other kinds of primary energy supplies create more precise forecasts by limiting price fluctuations?

Graph: International Energy Agency. Norway: Energy Mix. <https://www.iea.org/countries/norway/energy-mix>, accessed July 9, 2026.

Bundesnetzagentur. SMARD: Electricity and gas market data. <https://www.smard.de/home>, accessed June 23rd, 2026.

Eddy, M., & Alderman, L. (2022). "How Europeans Are Responding to Exorbitant Gas and Power Bills." The New York Times. https://www.nytimes.com/2022/02/21/business/europe-power-gas-bill.html?eafs_enabled=false, accessed June 28th, 2026.

Oltermann, P. (2022, July 19). Germany worries about gas rationing as supply from Russia halted. The Guardian. <https://www.theguardian.com/world/2022/jul/19/germany-worries-about-gas-rationing-as-supply-from-russia-halted>, accessed June 28th, 2026.

Statistisches Bundesamt. (2023, September 27). Stromverbrauch der privaten Haushalte nach Haushaltsgrößenklassen. <https://www.destatis.de/DE/Themen/Gesellschaft-Umwelt/Umwelt/UGR/private-haushalte/Tabellen/stromverbrauch-haushalte.html>

International Energy Agency. Norway: Energy Mix. <https://www.iea.org/countries/norway/energy-mix>, accessed July 9th, 2026.

International Energy Agency. Germany: Energy Mix. <https://www.iea.org/countries/germany/energy-mix>, accessed July 4th, 2026.